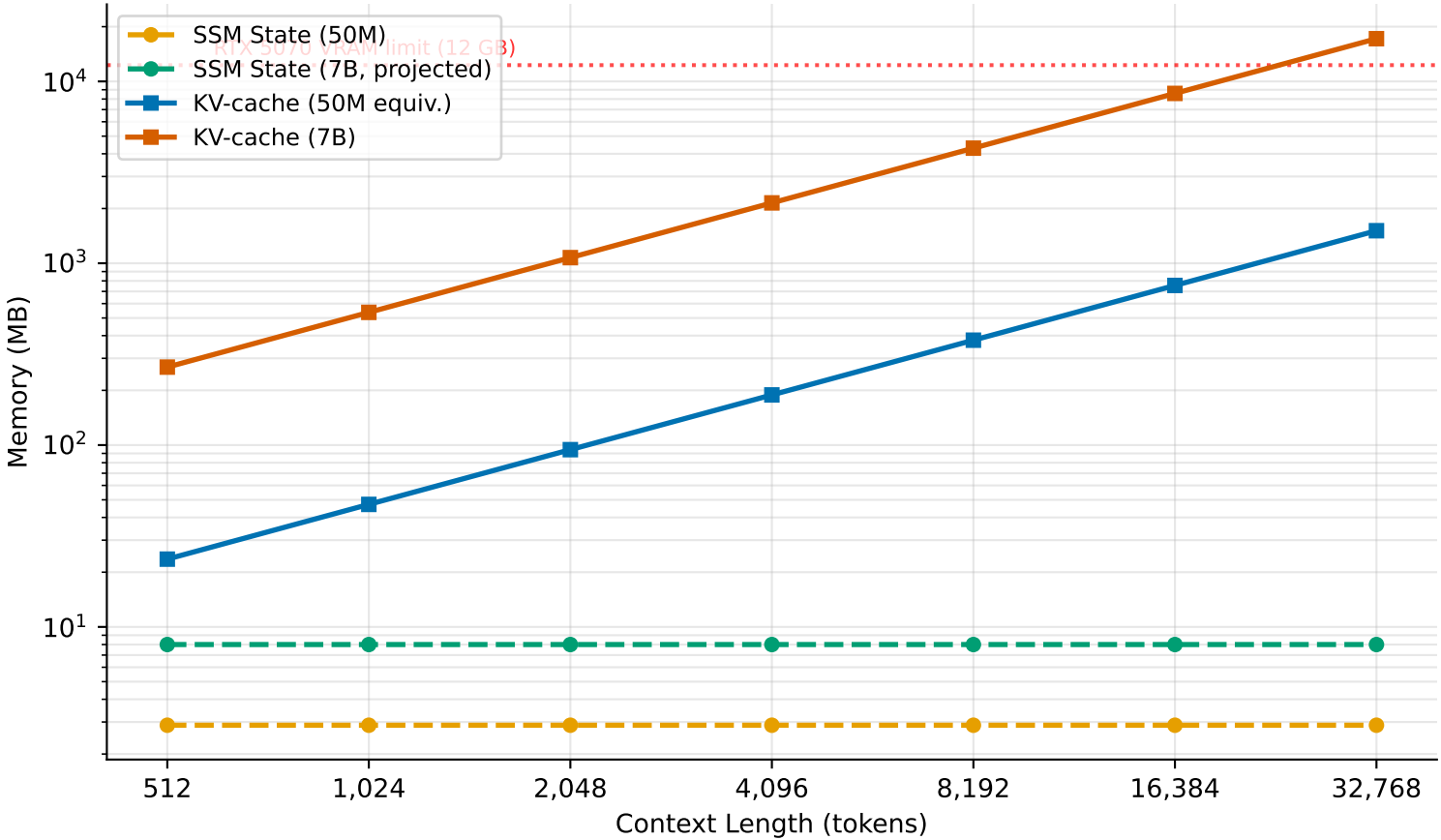
State Memory: SSM (O(1)) vs Transformer KV-cache (O(n))
SSM State (50M)
SSM State (7B, projected)
KV-cache (50M equiv.)
KV-cache (7B)
RTX 5070 VRAM limit (12 GB)
Memory (MB)
10^4
10^3
10^2
10^1
512
1,024
2,048
4,096
8,192
16,384
32,768
Context Length (tokens)